

Recording



CSC503 x SENG747: Data Mining
Fall 2026

Photo: Steve McCurry

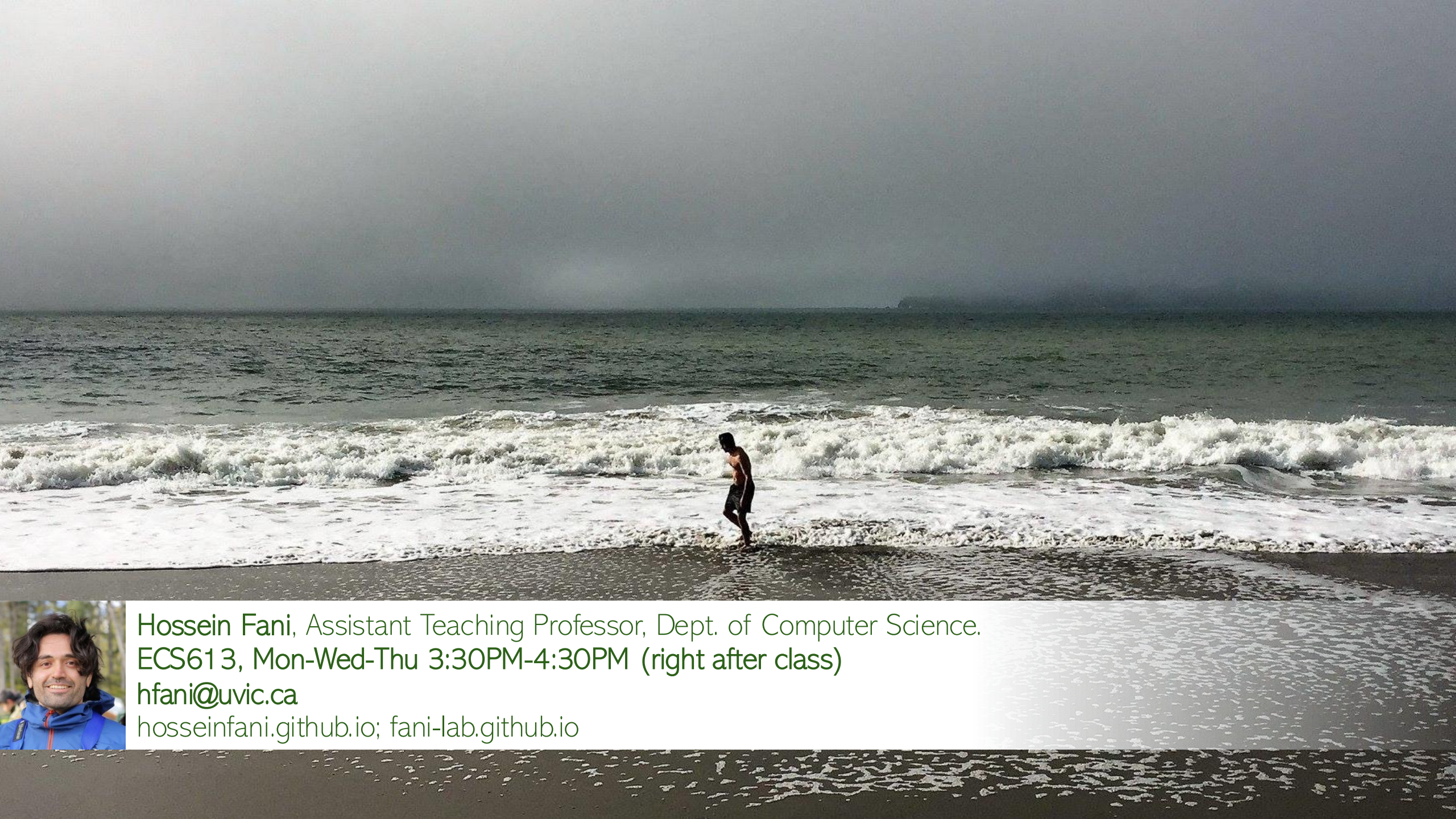


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Undergrad (80.8%) and Graduate (19.2%) Students

Undergraduate (97): Computer Science (61), Software Engineering (12), Psychology (6), Math (4), Data Science (3), Geomatics (2), Health Information Science (1), Non-Degree (3)

Graduate (23): Computer Science (8), Telecoms & Information Security (8), Applied Data Science (5), Electrical & Computer Engineering (1), Management (1)



Hossein Fani, Assistant Teaching Professor, Dept. of Computer Science.
ECS613, Mon-Wed-Thu 3:30PM-4:30PM (right after class)

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hosseinfani.github.io; fani-lab.github.io



The Constellation Fornax /'fɔːrnæks/

By Hubble Ultra Deep Field, observed 2003, yielding clues to the birth rate of stars over the 13-billion-year universe.
<https://hubblesite.org/contents/media/videos/2014/27/766-Video.html>





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Data Mining to World Model as Stars to Science

Stars may not be the first, but they are among humanity's foremost problems to solve. From the primitive problem of measuring distance to today's pursuit of World Models in AI.

Scoping (Localizing) Data Mining

What are we talking about when we talk about Data Mining?

Stay focused as we zoom in (drill down) through our odyssey ...

Outline: Week 01 A: Monday, B: Wednesday, C: Thursday

I) An Odyssey

II) Artificial Intelligence (AI)

III) State of AI

III) Data Mining

Outline: Week 02 A: Monday, Course Logistics & Topics

Outline: W01 B

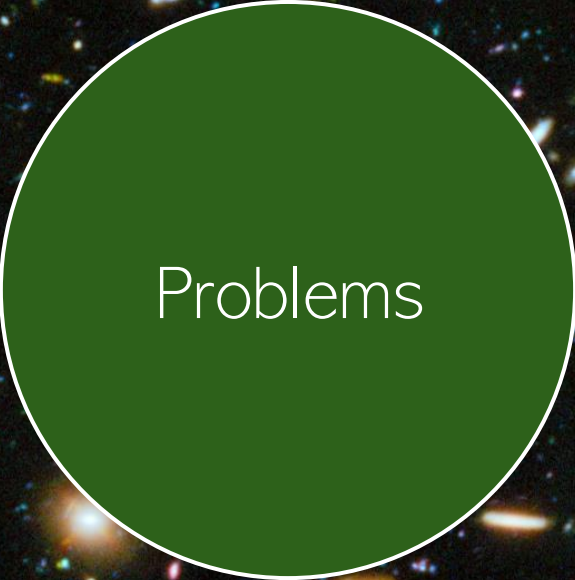
I) An Odyssey

II) Artificial Intelligence (AI)

III) State of AI

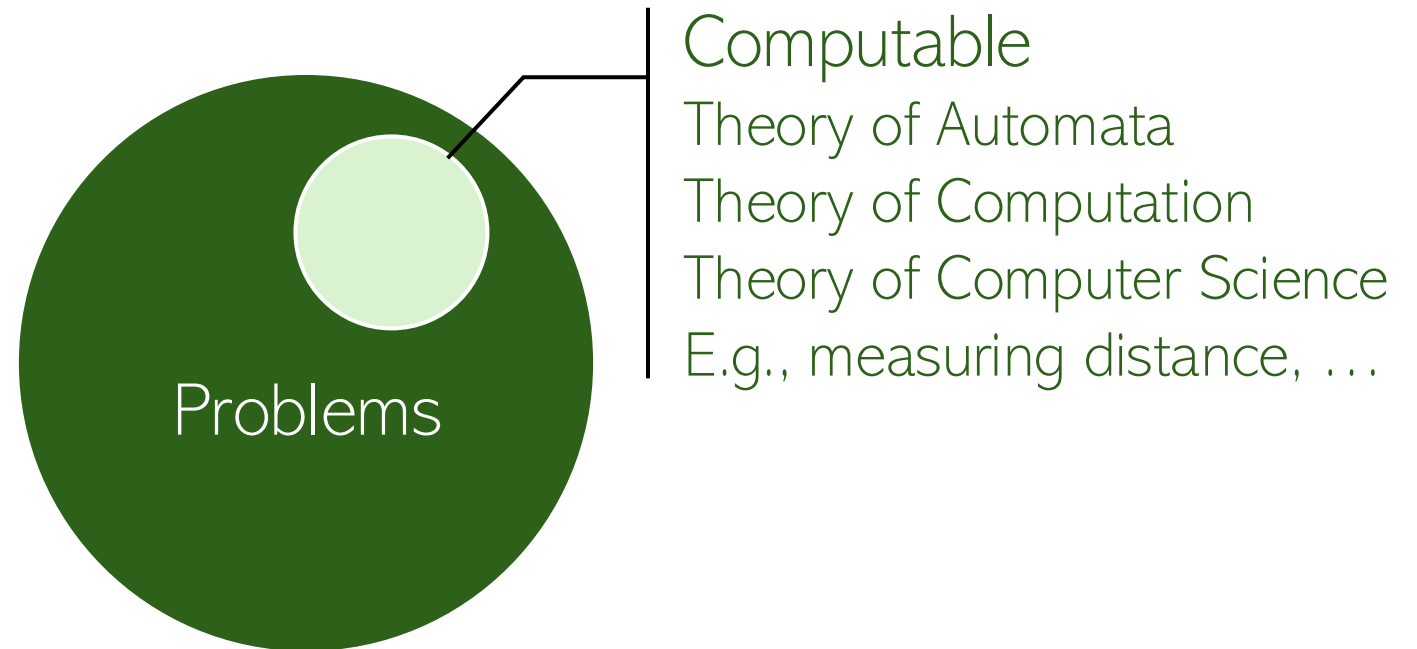
III) Data Mining

A Data Mining Odyssey



Problems

A Data Mining Odyssey

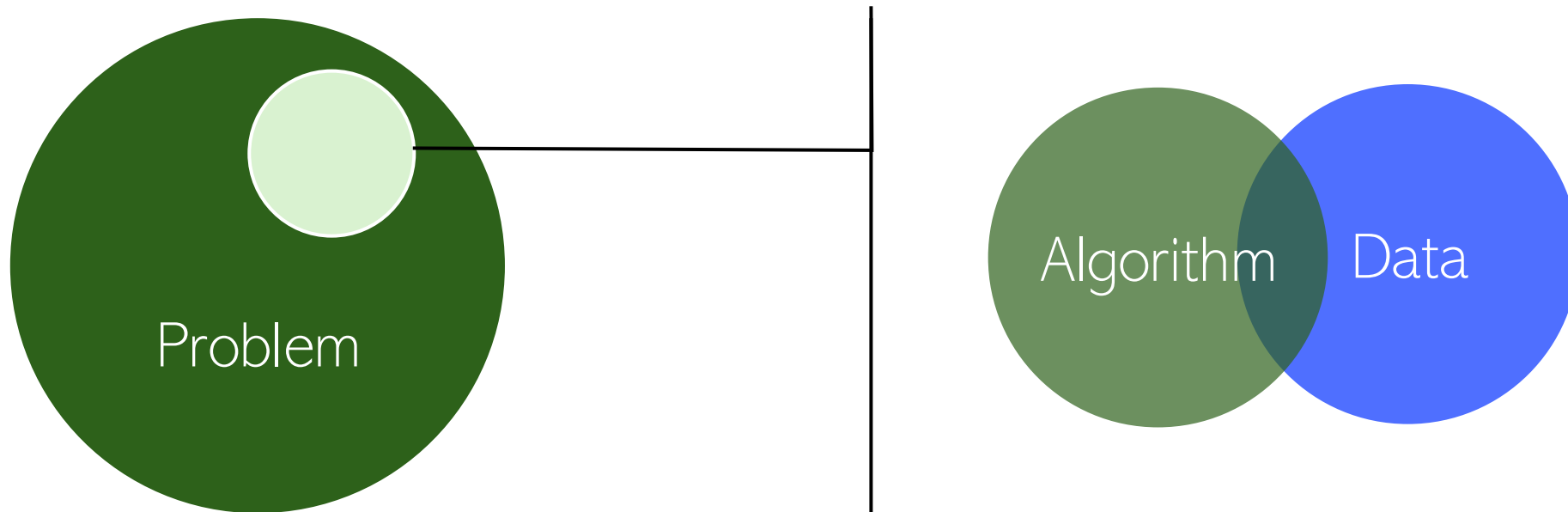


Incompleteness: Not all problems have solutions!

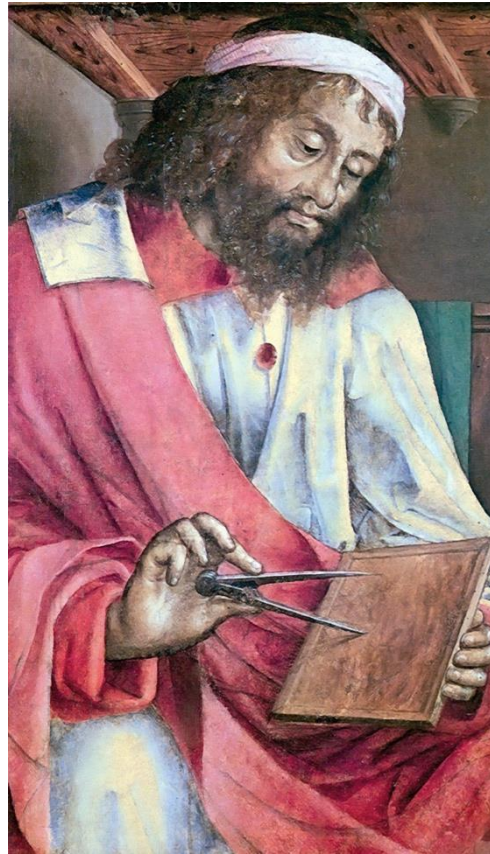
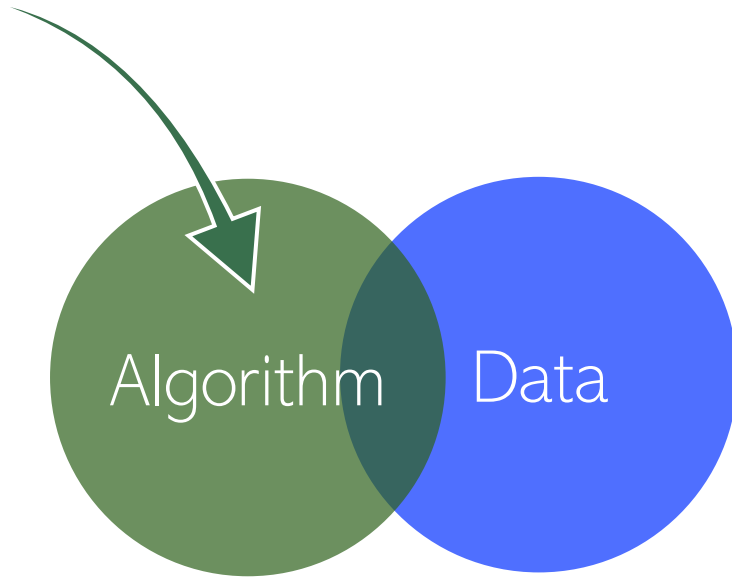
Undecidability: Even if there is a solution, we may not obtain it!

Complexity: Even if we can obtain it, it may take forever (NP)!

A Data Mining Odyssey



A Data Mining Odyssey



Muhammad ibn Musa al-Khwarizmi

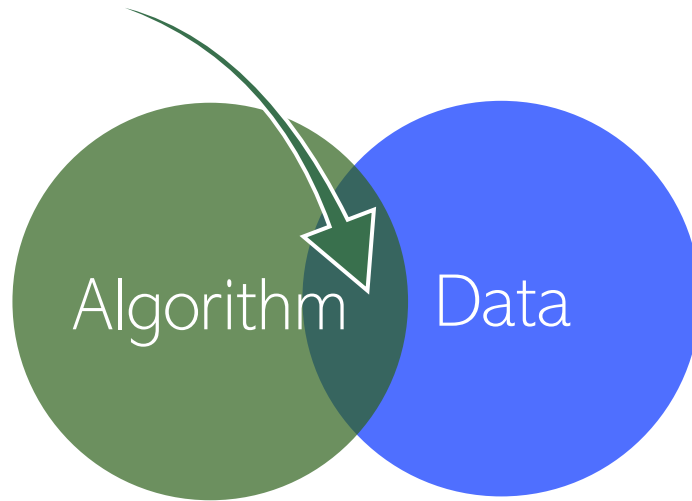
Worked on mathematics, astronomy, and geography. Widely influential in the Islamic world and Europe.

Euclid – 300 BCE
Greatest Common Divisor

Archimedes – 212 BCE
Approximate π

Khwārizmī الخوارزمي – 850 AD
Linear and Quadratic Equations

A Data Mining Odyssey



Data gets bigger!
Algorithms get slower!

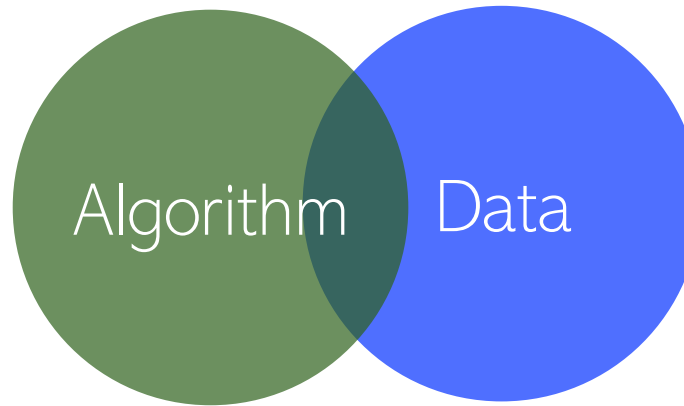
Data Engineers $\leftrightarrow$ Algorithm Designers

Better Data Modeling $\leftrightarrow$ Faster Algorithm

- Sorting in Arrays vs. Trees
- Searching in Arrays vs. Trees
- Linear Algebra on Sparse Matrix in Arrays vs. Linked Lists
- ...

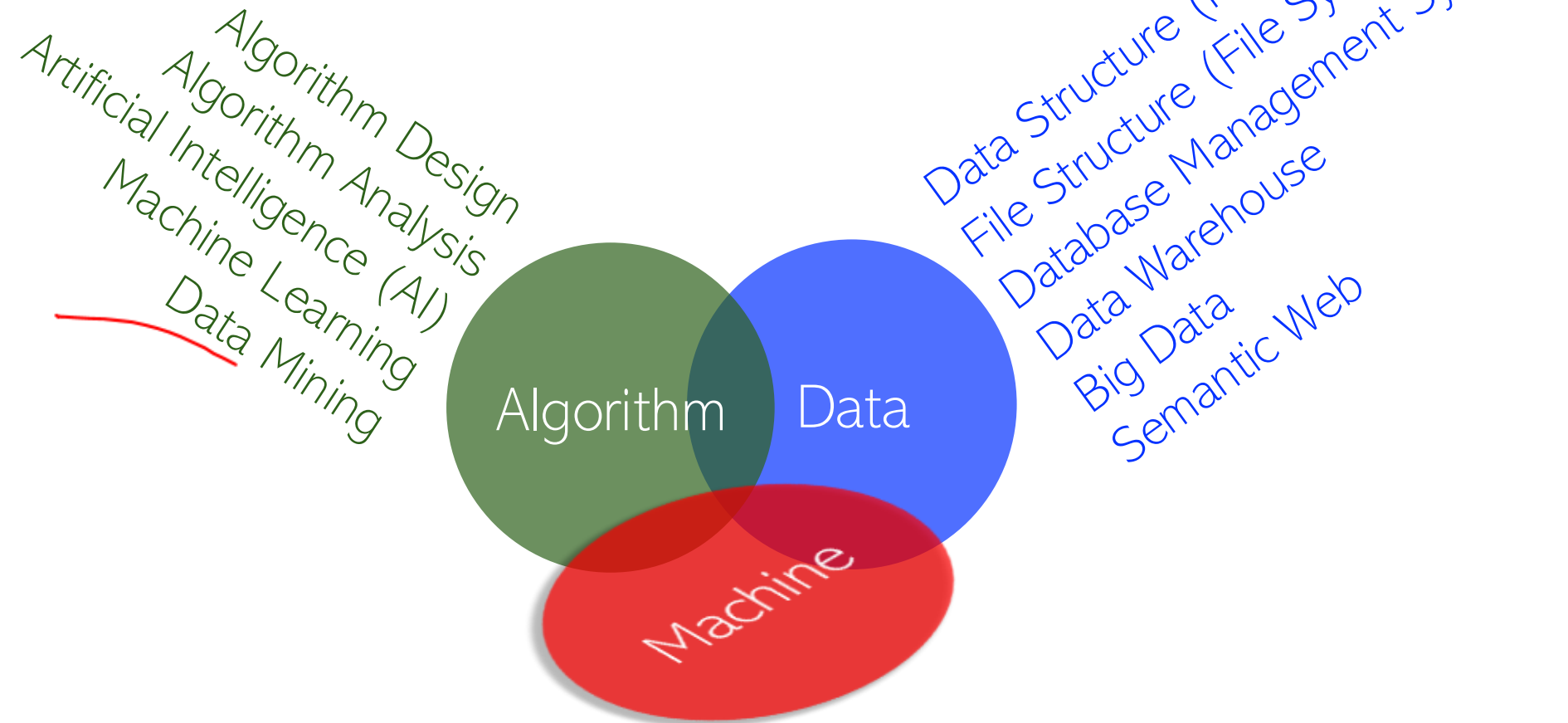
A Data Mining Odyssey

Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining



Data Structure (Memory)
File Structure (File Systems)
Database Management Systems
Data Warehouse
Big Data
Semantic Web

A Data Mining Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

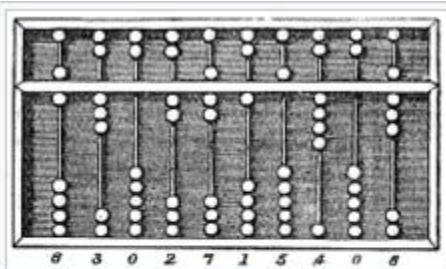
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A Data Mining Odyssey

<https://en.wikipedia.org/wiki/Computer#History>



The Ishango bone, a bone tool dating back to prehistoric Africa.



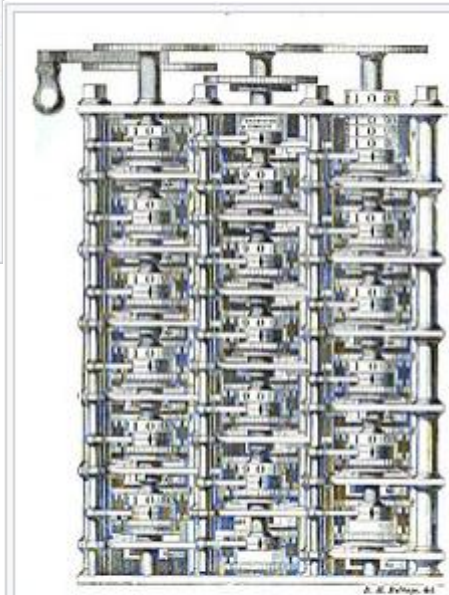
The Chinese **suanpan** (算盘). The number represented on this **abacus** is 6,302,715,408.

2700 – 2300 BC



The **Antikythera mechanism**, dating back to **ancient Greece** circa 150–100 BC, is an early **analog computing device**.

100 BC: to calculate astronomical positions



A portion of **Babbage's Difference engine**.

1833: Father of Computer

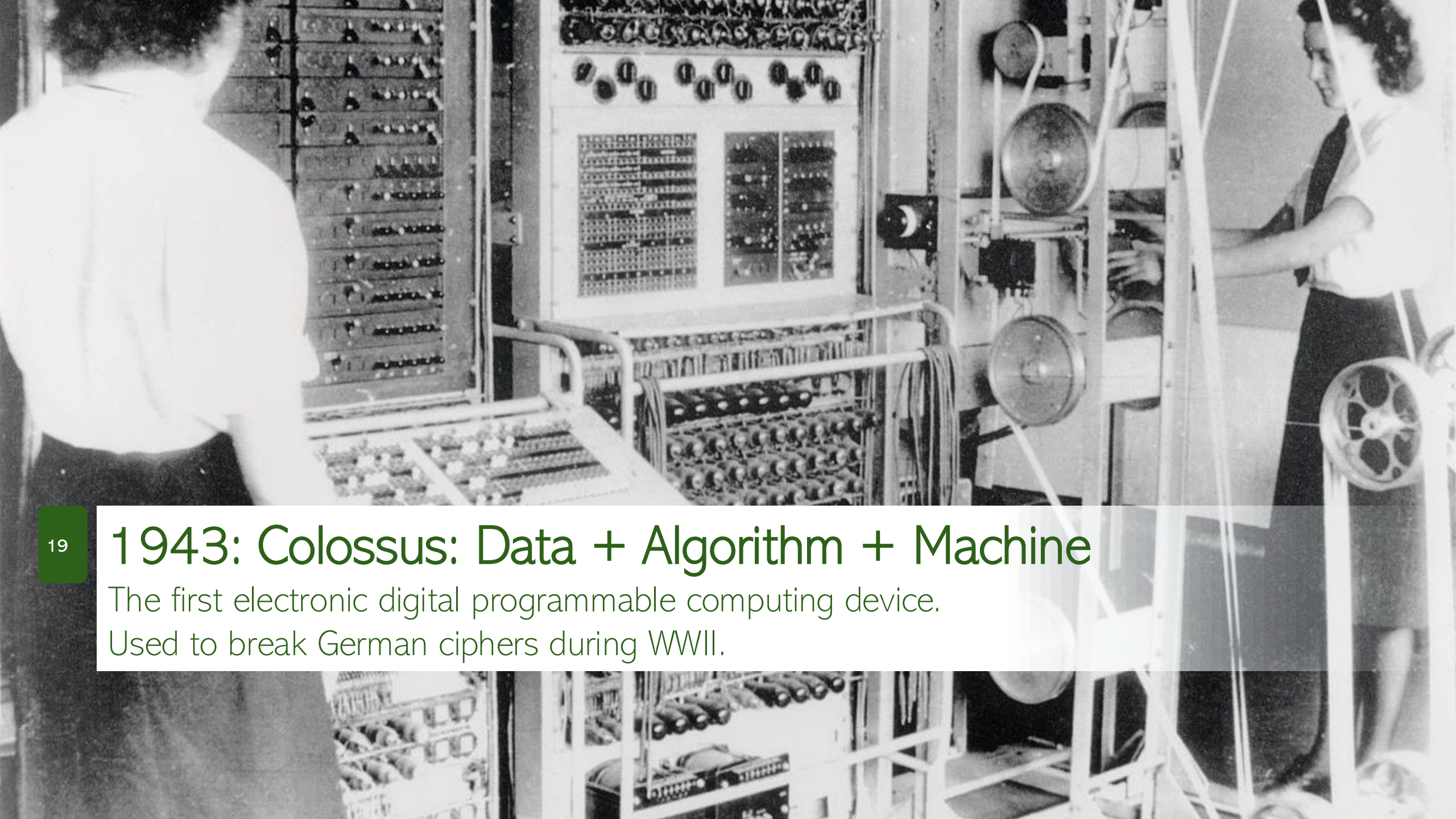


Replica of **Zuse's Z3**, the first fully automatic, digital (electromechanical) computer.

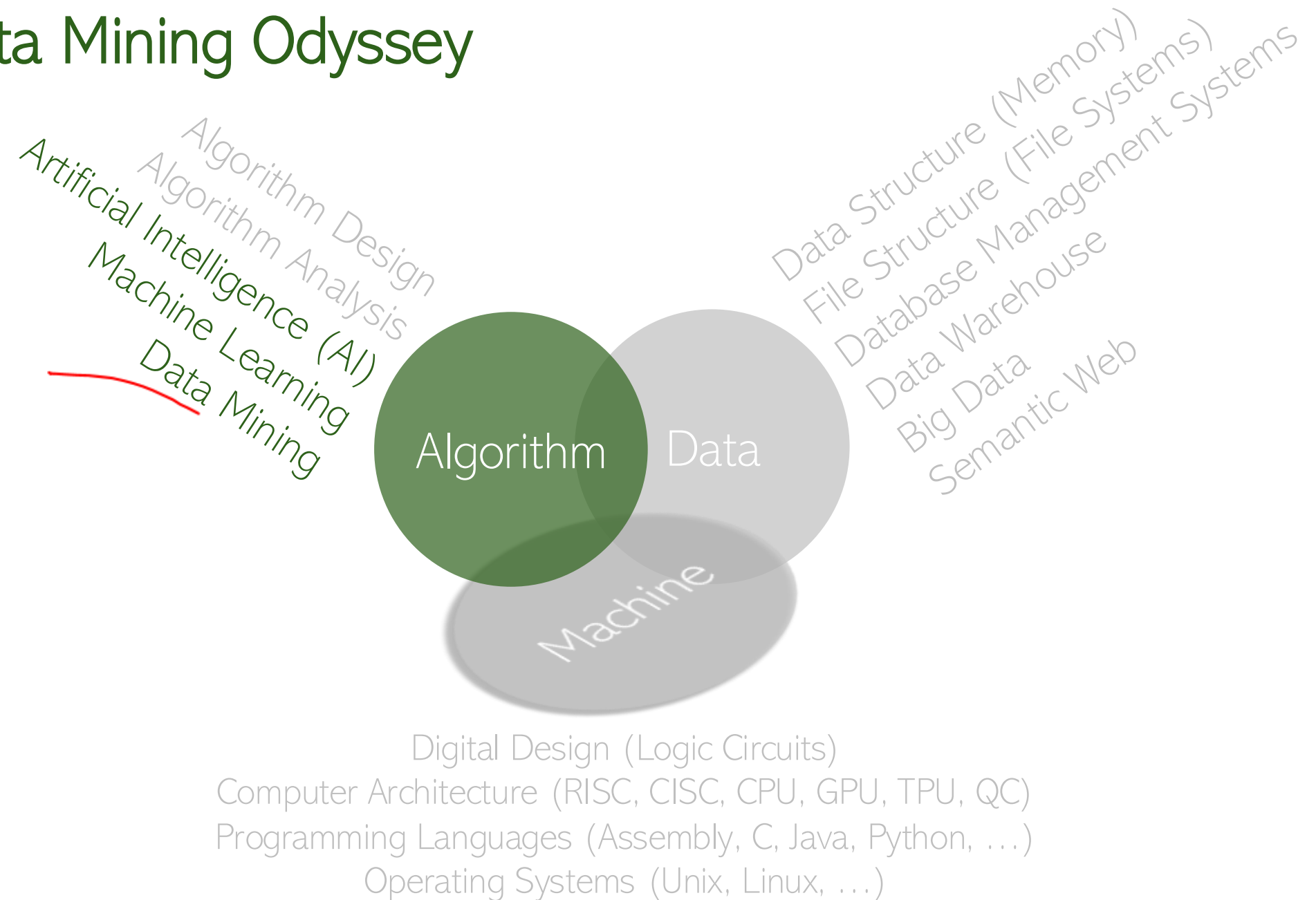
1941

1943: Colossus: Data + Algorithm + Machine

The first electronic digital programmable computing device.
Used to break German ciphers during WWII.



A Data Mining Odyssey



Outline: W01 B

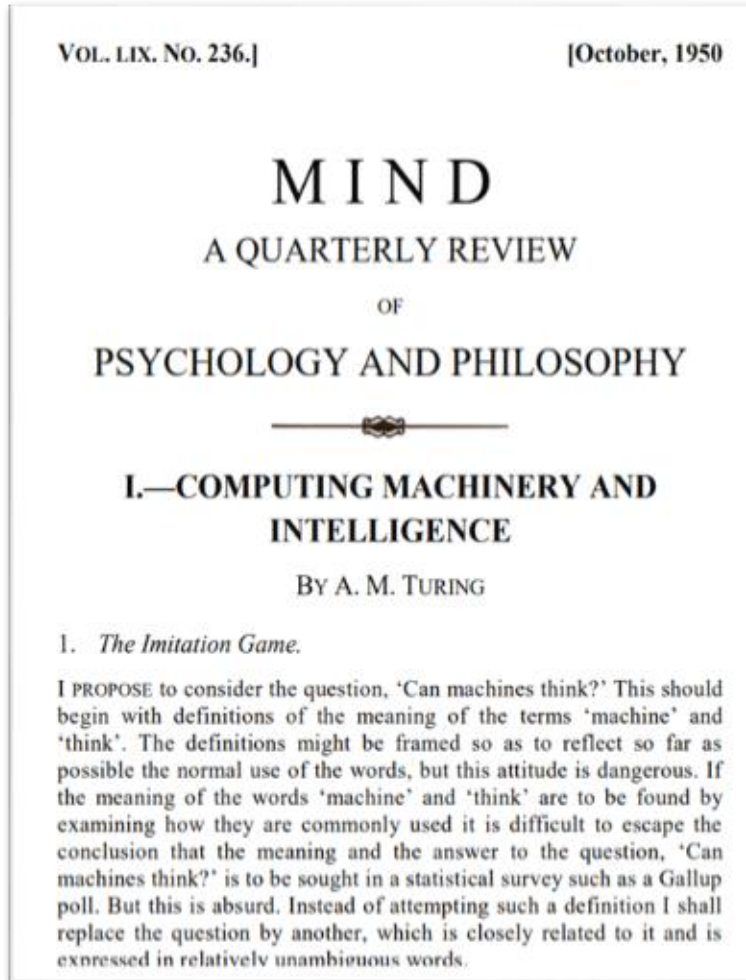
I) An Odyssey

II) Artificial Intelligence (AI)

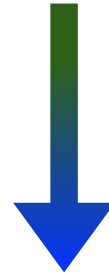
III) State of AI

III) Data Mining

Forms of Intelligence



Human-level



From Processing to Understanding to Communication
Text, Speech, Visual, Cognition, Reason

Intelligence = Emulate Human: Imitation Game

Alan Mathison Turing (/ˈtʃʊərɪŋ/; 1912 – 1954)

English mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist.

Widely considered to be the father of theoretical computer science.

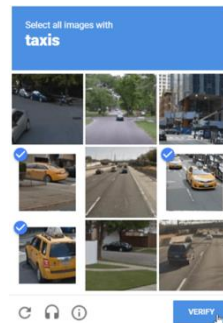
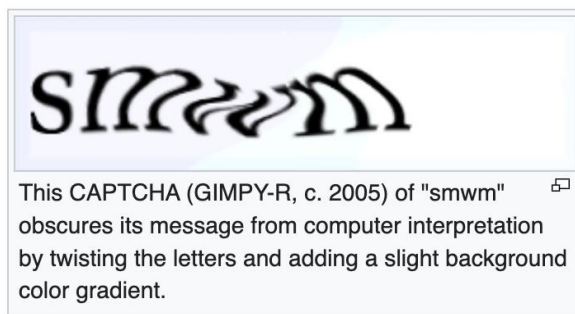
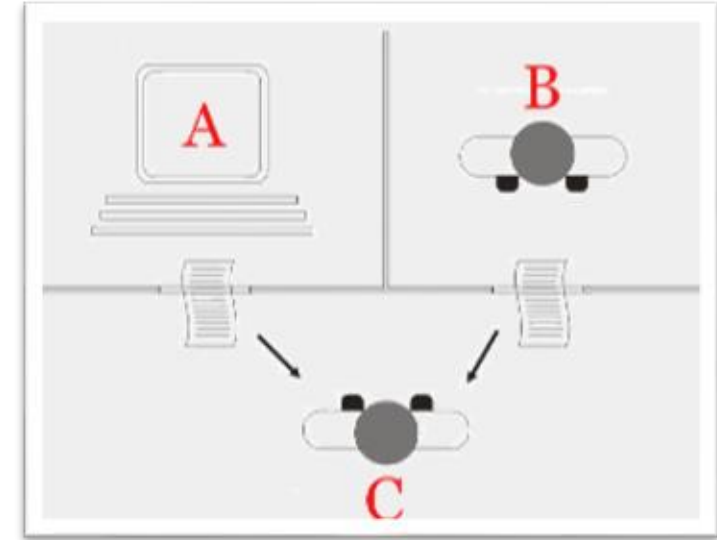
Influential in theoretical computer science, formalisation of the concepts of algorithm and computation with the Turing machine.



Artificial Intelligence: Turing Test

Human player C, the interrogator, is trying to determine which player – A xor B – is a computer and which is a human. The interrogator is limited to using the responses to written questions to make the determination. An intelligent machine is to fool the interrogator by exhibit equivalent to, or indistinguishable from, that of a human.

Fast vs. Slow? Wrong vs. Right? Inaccurate vs. Precise?
Limited to text but can be generalized to other modalities:
Speech? Body Language?



CAPTCHA (/ˈkæp.tʃə/ KAP-chə) To deter bot attacks.
Completely Automated Public Turing test to tell Computers and Humans Apart



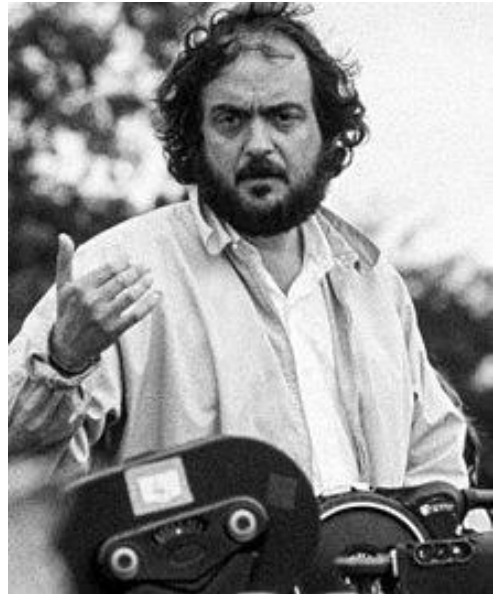
Artificial Intelligence: Turing Test

1968: A Conversation with HAL!

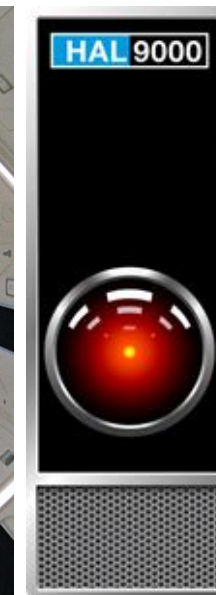
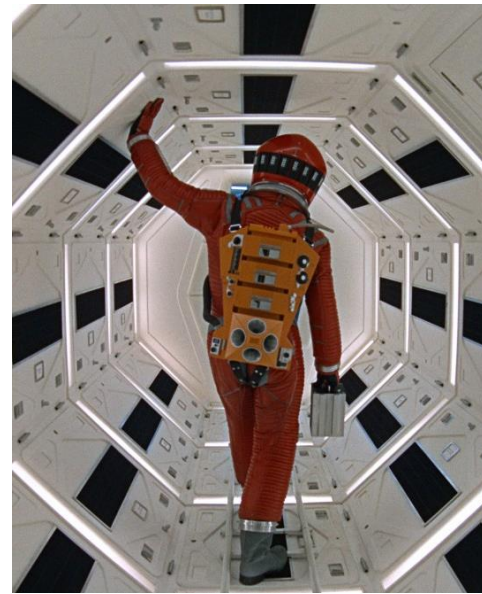
“2001: A Space Odyssey”, Stanley Kubrick: <https://www.youtube.com/watch?v=r13I-TuDcWI>



Sir Arthur Charles Clarke (1917 – 2008)
English science-fiction writer,
science, writer, futurist, inventor,
undersea explorer, and tv series host.



Stanley Kubrick (1928 – 1999)
American filmmaker and photographer.
Regarded as one of the greatest and most influential filmmakers.
His films gained recognition for their attention to detail,
innovative cinematography, extensive set design, and dark humor.



HAL:
Heuristically program
ALgorithmic computer

Controls the
Discovery One spacecraft and
interacts with the
ship's astronaut crew.
https://en.wikipedia.org/wiki/HAL_9000

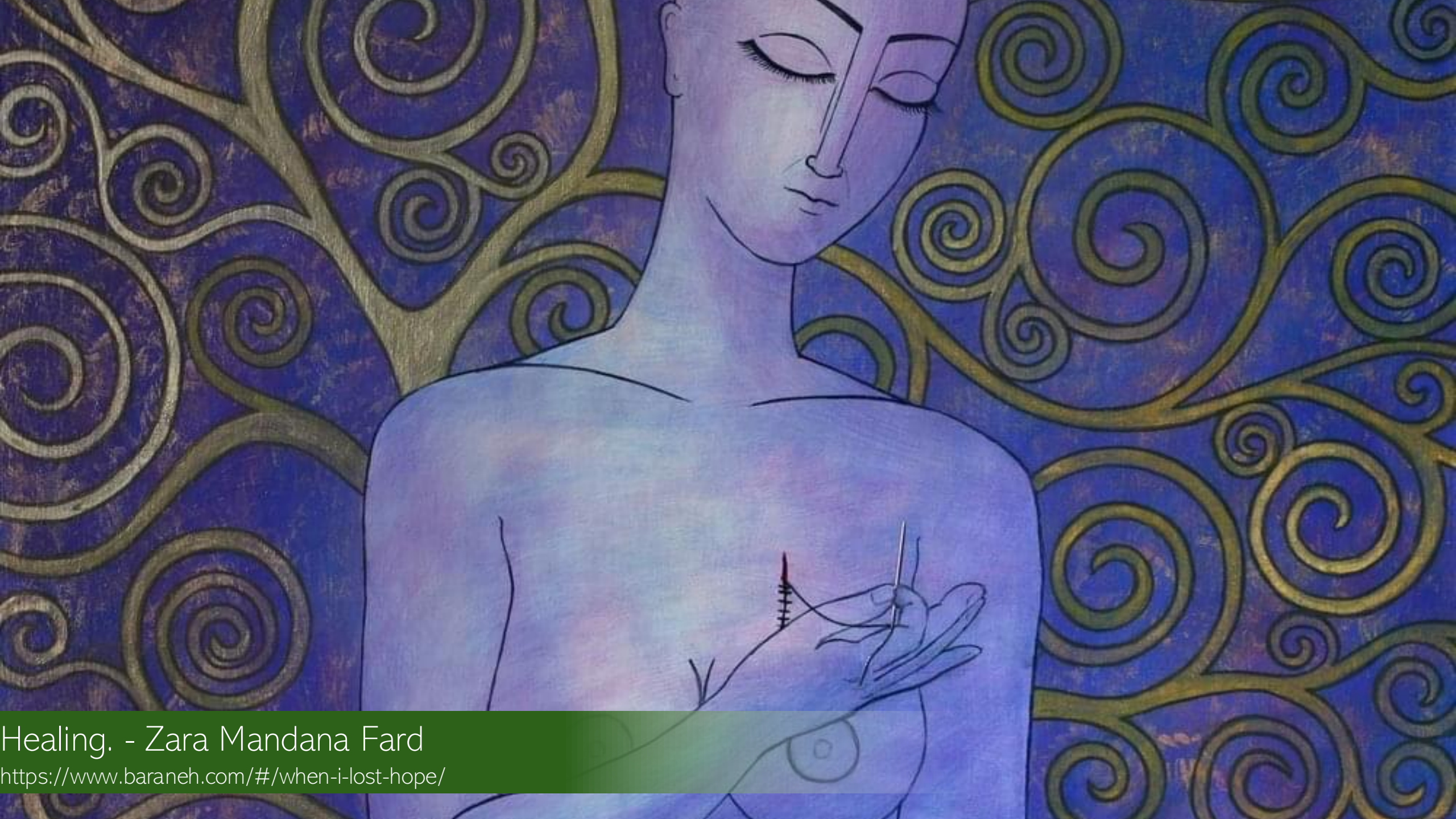
HD





Look! - Alex Andreev

<https://alexandreev.com/collections/free-downloads/look>



Healing. - Zara Mandana Fard

<https://www.baraneh.com/#/when-i-lost-hope/>

Artificial Intelligence: Turing Test + Emotion

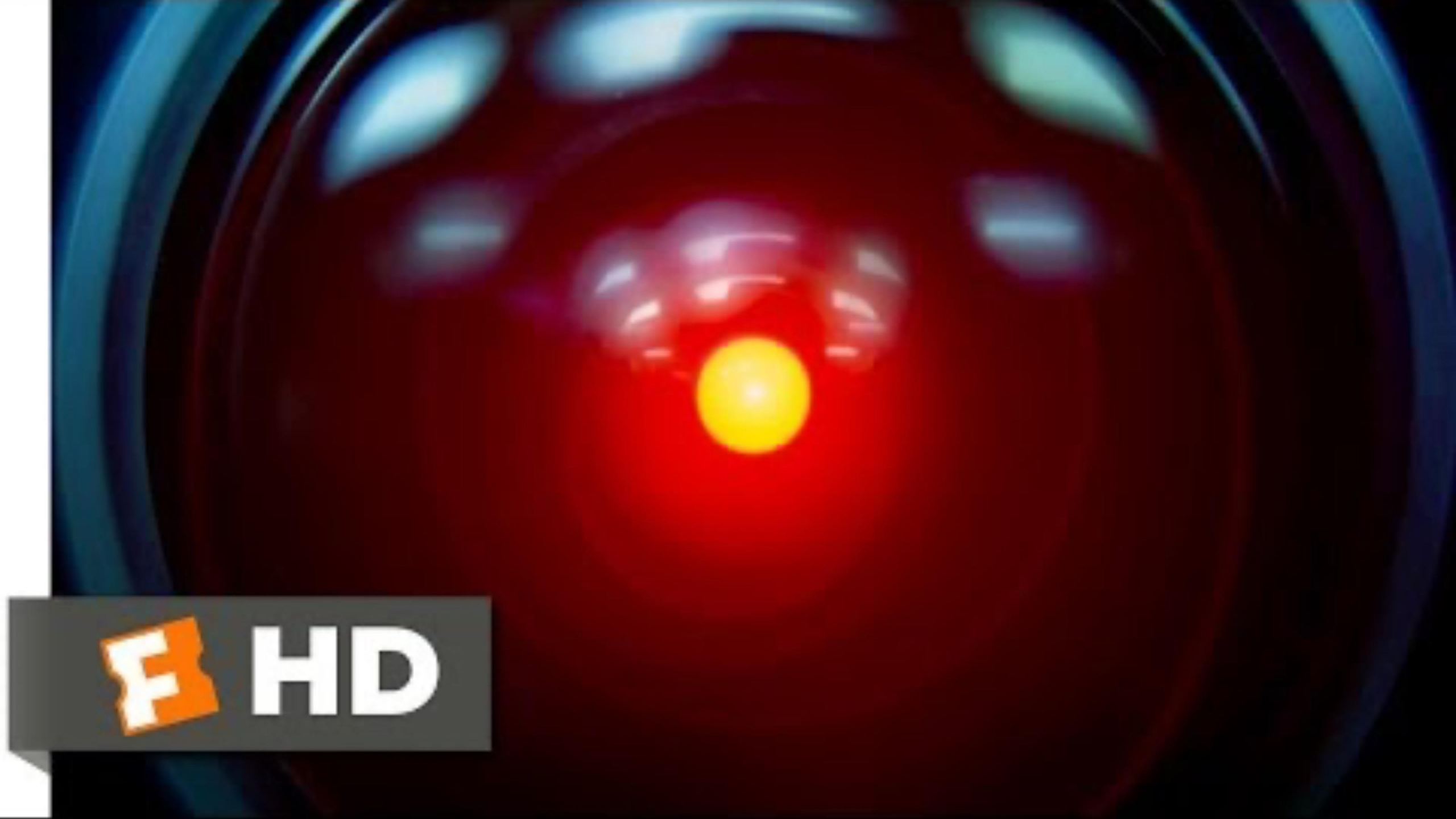
HAL: I'm Sorry, Dave! <https://www.youtube.com/watch?v=Wy4EfdnMZ5g>

Sentient /'sen.ʃəns/ or /'sen.ti.əns/ AI
Capacity to experience/express feelings.

Artificial Intelligence: Turing Test + Cognition

I'm afraid I cannot do that! This mission is too important for me to allow you to jeopardize it.
<https://www.youtube.com/watch?v=Wy4EfdnMZ5g>





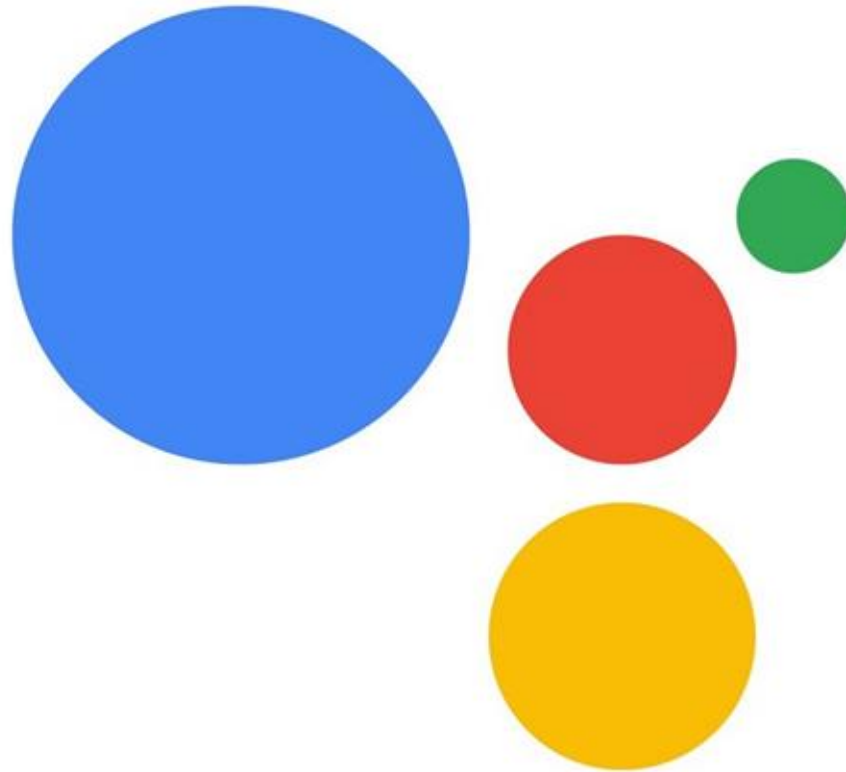
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Artificial Intelligence: Turing Test

2018: A Conversation with Google Duplex!

Sundar Pichai, Google's CEO: <https://www.youtube.com/watch?v=D5VN56jQMWM>

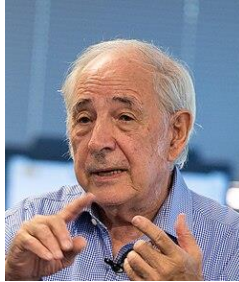




Artificial Intelligence: Turing Test is Misleading

*"The ability to **fake intelligence** does not amount to having intelligence."*

John Searle, Philosopher, "Chinese Room" argument to challenge Turing's imitation game, 1980.



*"Human-Level Intelligence or **Animal-Like Abilities**?"*

Adnan Darwiche, University of California.



On the Dangers of **Stochastic Parrots**: Can Language Models Be Too Big? 🦜

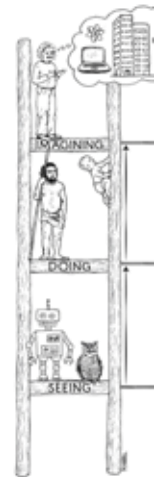
Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, FAccT 2021



The Book of **Why**: The New Science of **Cause and Effect**

The Ladder of Causation

Judea Pearl, 2018, Turing Award Winner



Artificial Intelligence: <https://openreview.net/forum?id=uyTL5Bvosj>

Published in Transactions on Machine Learning Research (05/2023)

Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models

Alphabetic author list:*

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askell, Amanda Dsouza, Ambrose Slone, Ameet Rahane, Anantharaman S. Iyer, Anders Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Ghahramani, Arif Tabassum, Arul Menezes, Arun Kirubakaran, Asher Mullokkandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut Erdem, Ayta Karakas, B. Ryan Roberts, Bao Sheng Loo, Barret Zoph, Bartłomiej Bojanowski, Batuhan Özyurt, Behnam Hedayatnia, Behnam Neyshabur, Benjamin Inden, Benno Stein, Berk Ekmekci, Bill Yuchen Lin, Blake Howald, Bryan Orinon, Cameron Diao, Cameron Dour, Catherine Stinson, Cedrick Argueta, César Ferri Ramírez, Chandan Singh, Charles Rathkopf, Chenlin Meng, Chitta Baral, Chiyu Wu, Chris Callison-Burch, Chris Waites, Christian Voigt, Christopher D. Manning, Christopher Potts, Cindy Ramirez, Clara E. Rivera, Clemencia Siro, Colin Raffel, Courtney Ashcraft, Cristina Garbacea, Damien Sileo, Dan Garrette, Dan Hendrycks, Dan Kilman, Dan Roth, Daniel Freeman, Daniel Khashabi, Daniel Levy, Daniel Moseguí González, Danielle Perszyk, Danny Hernandez, Danqi Chen, Daphne Ippolito, Dar Gilboa, David Dohan, David Drakard, David Jurgens, Debajyoti Datta, Deep Ganguli, Denis Emelin, Denis Kleyko, Deniz Yuret, Derek Chen, Derek Tam, Dieuwke Hupkes, Diganta Misra, Dilyar Buzan, Dimitri Coelho Mollo, Diyi Yang, Dong-Ho Lee, Dylan Schrader, Ekaterina Shutova, Ekin Dogus Cubuk, Elad Segal, Eleanor Hagerman, Elizabeth Barnes, Elizabeth Donoway, Ellie Pavlick, Emanuele Rodola, Emma Lam, Eric Chu, Eric Tang, Erkut Erdem, Ernie Chang, Ethan A. Chi, Ethan Dyer, Ethan Jerzak, Ethan Kim, Eunice Engfu Manyasi, Evgenii Zheltonozhskii, Fanyue Xia, Fatemeh Siar, Fernando Martínez-Plumed, Francesca Happé, François Chollet, Frieda Rong, Gaurav Mishra, Genta Indra Winata, Gerard de Melo, Germán Kruszewski, Giambattista Parascandolo, Giorgio Mariani, Gloria Wang, Gonzalo Jaimovitch-López, Gregor Betz, Guy Gur-Ari, Hana Galijasevic, Hannah Kim, Hannah Rashkin, Hannaneh Hajishirzi, Harsh Mehta, Hayden Bogar, Henry Shevlin, Hinrich Schütze, Hiromu Yakura, Hongming Zhang, Hugh Mee Wong, Ian Ng, Isaac Noble, Jaap Jumelet, Jack Geissinger, Jackson Kernion, Jacob Hilton, Jaehoon Lee, Jaime Fernández Fisac, James B. Simon, James Koppel, James Zheng, James Zou, Jan Kocoń, Jana Thompson, Janelle Wingfield, Jared Kaplan, Jarema Radom, Jascha Sohl-Dickstein, Jason Phang, Jason Wei, Jason Yosinski, Jekaterina Novikova, Jelle Bosscher, Jennifer Marsh, Jeremy Kim, Jeroen Taal, Jesse Engel, Jesujoba Alabi, Jiacheng Xu, Jiaming Song, Jillian Tang, Joan Waweru, John Burden, John Miller, John U. Baw, Jonathan Batchelder, Jonathan Berant, Jörg Froberg, Jose Rozen, Jose Hernandez-Orallo, Joseph Boudeman, Joseph Guerr, Joseph Jones, Joshua B. Tenenbaum, Joshua S. Rule, Joyce Chua, Kamil Kancierz, Karen Livescu, Karl Krauth, Karthik Gopalakrishnan, Katerina Ignatyeva, Katja Markert, Kaustubh D. Dhole, Kevin Gimpel, Kevin Omondi, Kory Mathewson, Kristen Chiafullo, Ksenia Shkaruta, Kumar Shridhar, Kyle McDonell, Kyle Richardson, Laria Reynolds, Leo Gao, Li Zhang, Liam Dugan, Lianhui Qin, Lidia Contreras-Ochando, Louis-Philippe Morency, Luca Moschella, Lucas Lam, Lucy Noble, Ludwig Schmidt, Luheng He, Luis Oliveros Colón, Luke Metz, Lütfi Kerem Şenel, Maarten Bosma, Maarten Sap, Maartje ter Hoeve, Maheen Farooqi, Manaal Faruqi, Mantas Mazeika, Marco Baturan, Marco Marelli, Marco Maru, Maria Jose Ramírez Quintana, Marie Tolkiehn, Mario Giulianelli, Martha Lewis, Martin Pot-thast, Matthew L. Leavitt, Matthias Hagen, Mátyás Schubert, Medina Orduna Baiteirova, Melody Arnaud, Melvin McElrath, Michael A. Yee, Michael Cohen, Michael Gu, Michael Ivanitskiy, Michael Starritt, Michael Strube, Michal Swędrowski, Michele Bevilacqua, Michihiro Yasunaga, Mihir Kale, Mike Cain, Mimese Xu, Mirac Suzgun, Mitch Walker, Mo Tiwar, Mohit Bansal, Moin Aminnaseri, Mor Geva, Mozdeh Ghahramani, Mukund Varma T, Nanyun Peng, Nathan A. Chi, Nayeon Lee, Neta Gur-Ari Krakover, Nicholas Cameron, Nicholas Roberts, Nick Doiron, Nicole Martinez, Nikita Nangia, Niklas Deckers, Niklas Muennighoff, Nitish Shirish Keskar, Niveditha S. Iyer, Noah Constant, Noah Fiedel, Nuan Wen, Oliver Zhang, Omar Agha, Omar Elbaghdadi, Omer Levy, Owain Evans, Pablo Antonio Moreno Casares, Parth Doshi, Pascale Fung, Paul Pu Liang, Paul Vicol, Pegah Alipoormolabashi, Pelyuan Liao, Percy Liang, Peter Chang, Peter Eckersley, Phu Mon Htut, Pinyu Hwang, Piotr Milkowski, Piyyush Patil, Pouya Pezeshkpour, Priti Oli, Qiaozhu Mei, Qing Lyu, Qinqin Chen, Rabin Banjade, Rachel Etta Rudolph, Raefer Gabriel, Rahel Habacker, Ramon Risco, Raphaël Millière, Rhythm Garg, Richard Barnes, Rif A. Saurous, Riku Arakawa, Robbe Raymaekers, Robert Frank, Rohan Sikand, Roman Novak, Roman Shtelev, Ronan LeBras, Rosanne Liu, Rowan Jacobs, Rui Zhang, Ruslan Selakutdinov, Ryan Chi, Ryan Lee, Ryan Stovall, Ryan Yang, Sahib Singh, Saif M. Moham-mad, Sajant Anand, Sam Dillavou, Sam Shleifer, Sam Wiseman, Samuel Gruetter, Samuel R. Bowman, Samuel S. Schoenholz, Sanghyun Han, Sanjeev Kwatra, Sarah A. Rous, Sarik Ghazarian, Sayan Ghosh, Sean Casey, Sebastian Bischoff, Sebastian Gehrmann, Sebastian Schuster, Sepideh Sadeghi, Shadi Hamdan, Sharon Zhou, Shashank Srivas-tava, Sherry Shi, Shikhar Singh, Shima Asaadi, Shixiang Shane Gu, Shubh Pachchigar, Shubham Toshniwal, Shyam Upadhyay, Shyamolima (Shammie) Debnath, Siamak Shakeri, Simon Thormeyer, Simone Melzi, Siva Reddy, Sneha Priscilla Makini, Soo-Hwan Lee, Spencer Torene, Sriharsha Hatwar, Stanislas Dehaene, Stefan Dovic, Stefano Ermon, Stella Biderman, Stephanie Lin, Stephen Prasad, Steven T. Piantadosi, Stuart M. Shieber, Summer Misherghi, Svetlana Kiritchenko, Swaroop Mishra, Tal Linzen, Tal Schuster, Tao Li, Tao Yu, Tariq Ali, Tatsuo Hashimoto, Te-Lin Wu, Théo Desbordes, Theodore Rothschild, Thomas Phan, Tianle Wang, Tiberius Nkinyili, Timo Schick, Timofei Kornev, Titus Dumny, Tobias Gerstenberg, Trenton Chang, Trishala Neeraj, Tushar Khot, Tyler Shultz, Uri Shaham, Vedant Misra, Vera Demberg, Victoria Nyamai, Vikas Raunak, Vinay Ramasesh, Vinay Uday Prabhu, Vishakh Padmakumar, Vivek Srikanar, William Fedus, William Saunders, William Zhang, Wout Vossen, Xiang Ren, Xiaoyu Tong, Xinran Zhao, Xinyi Wu, Xudong Shen, Yaddollah Yaghoobzadeh, Yair Lakretz, Yangqiu Song, Yasaman Bahri, Yejin Choi, Yichi Yang, Yiding Hao, Yifu Chen, Yonatan Belinkov, Yu Hou, Yufang Hou, Yuntao Bai, Zachary Seid, Zhuoye Zhao, Zijian Wang, Zijie J. Wang, Zirui Wang, Ziyi Wu

Reviewed on OpenReview: <https://openreview.net/forum?id=uyTL5Bvosj>

442 vs. 1



VOL. LIX, No. 236.]

[October, 1959]

MIND A QUARTERLY REVIEW OF PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

Artificial Intelligence: Emulate vs. Exceed Human



Yann André Le Cun (/lɑ̃ˈkʌn/ lə-KUN) (1960 - Present)

French-American computer scientist in the fields of AI, machine learning, computer vision, robotics and image compression.

Professor at the at New York University.

Served as Chief AI Scientist at Meta Platforms.

Language is an imperfect, incomplete, and low-bandwidth serialization protocol for the internal data structures we call thoughts.

9:36 AM · Mar 6, 2021 · Twitter for Android

“
Despite its many practical applications, language is perhaps number **300** in the priority list for AI research. It would be a great achievement if AI could attain the capabilities of an **orangutan**, which do not include language!

Artificial Intelligence: Exceed Human

HAL Reads Lips: <https://www.youtube.com/watch?v=XD08OYnmkNY>



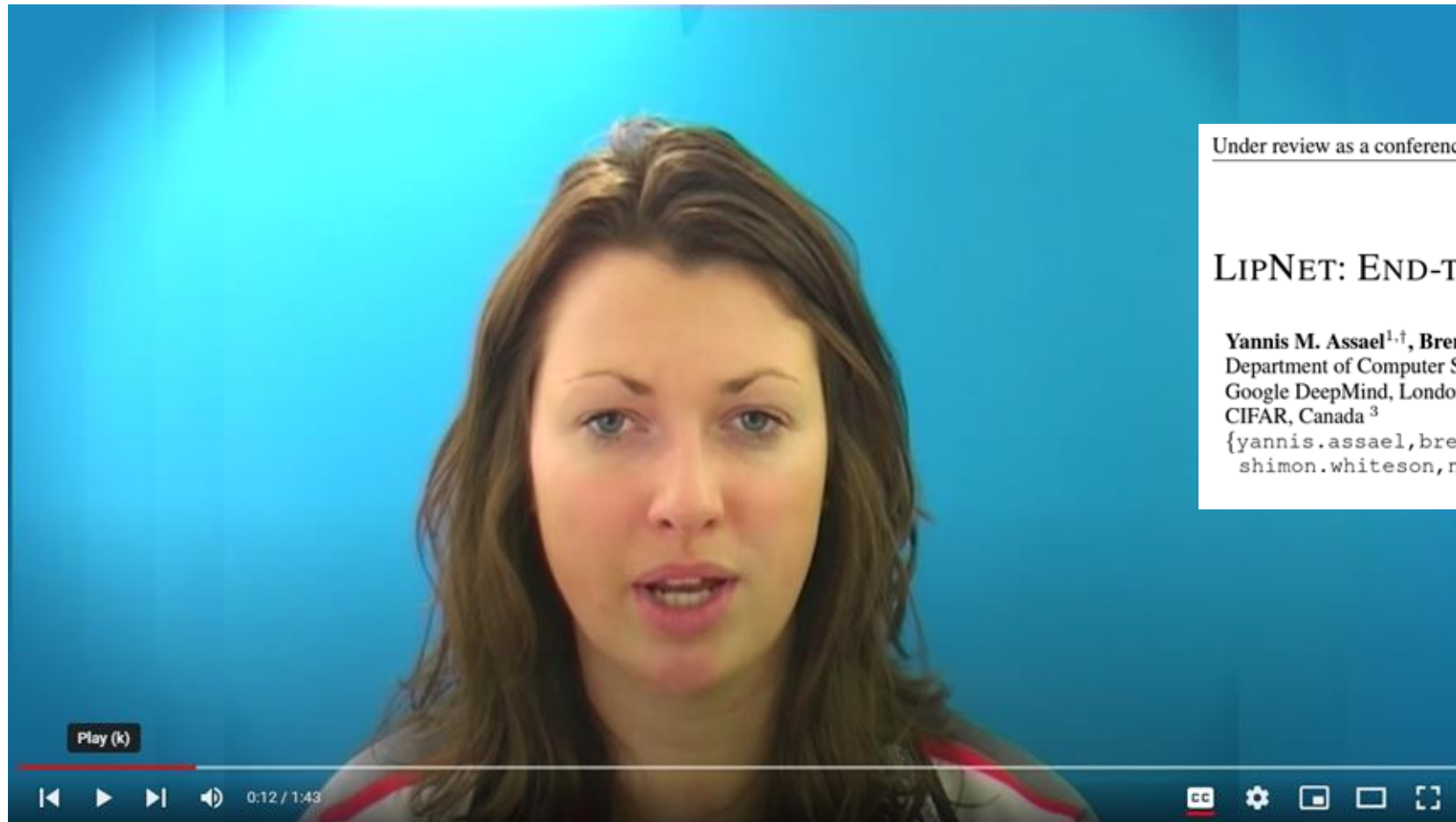


F HD

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Artificial Intelligence: Exceed Human Processing

<https://www.youtube.com/watch?v=fa5QGremQf8&list=PLXkuFIFnXUAPlrXKgtlpctv2NuSo7xw3k&index=1>



Under review as a conference paper at ICLR 2017

LIPNET: END-TO-END SENTENCE-LEVEL LIPREADING

Yannis M. Assael^{1,†}, Brendan Shillingford^{1,†}, Shimon Whiteson¹ & Nando de Freitas^{1,2,3}
Department of Computer Science, University of Oxford, Oxford, UK ¹
Google DeepMind, London, UK ²
CIFAR, Canada ³
{yannis.assael,brendan.shillingford,
shimon.whiteson,nando.de.freitas}@cs.ox.ac.uk

LipNet: How easy do you think lipreading is?

LipNet



Sentence: Place blue in m 1 soon
LipNet:

Hyperlinks: W01 B

Constellation Fornax: <https://science.nasa.gov/asset/hubble/hubble-unveils-its-most-colorful-view-of-the-universe-zoom-and-pan/>

A Conversation with HAL: <https://www.youtube.com/watch?v=r13I-TuDcWI>

I'm Sorry, Dave: <https://www.youtube.com/watch?v=Wy4EfdnMZ5g>

Google Duplex: <https://www.youtube.com/watch?v=D5VN56jQMWM>

Beyond the Imitation Game: <https://openreview.net/challenge?redirect=%2Fforum%3Fid%3DuyTL5Bvosj>

Hal Reads Lips: <https://www.youtube.com/watch?v=XDO8OYnmkNY>

LipNet: <https://www.youtube.com/watch?v=fa5QGremQf8&list=PLXkuFIFnXUAPlrXKgtIpctv2NuSo7xw3k&index=2>